

## Performance Work Statement (PWS)

### FY26 Spillway Drainage Inspection Garrison Dam, ND

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#### 1. Background

##### 1.1 PWS Summary and Purpose

This Performance Work Statement describes a scope of work and requirements for cleanout and camera inspection of the Garrison Dam spillway subdrains, the spillway adjacent box drains, the spillway crest relief wells, and the powerhouse under seepage drains. The objective of this project is to gain a better understanding of the condition of these features, informing design of future dam safety modifications, and to rehabilitate them as effectively as possible to maintain project performance and safety.

For the purpose of defining price schedule and performance requirements, the work has been divided into 6 tasks:

- Task 1 - Flush, measure sediment, camera and report for Rows 2-9 spillway subdrain (14,060 ft)
- Task 2 – Flush, measure sediment, camera and report for Row 1 spillway subdrain (2,120 ft)
- Task 3 – Flush, measure sediment, camera and report for spillway box drains (4,800 feet)
- Task 4 – Sounding, camera and report for crest relief wells
- Task 5 – Temporary chute access staircase
- Task 6 – All Other Work
- Task 7 (Option 1) - Flush, measure sediment, camera and report for power plant drainage (5,295 feet)

Tasks 1 through 4 are located around the Garrison Dam spillway at the dam's east abutment, while Task 7 is located outside of the project powerhouse near the west abutment. An overview map is provided in Figure 1.

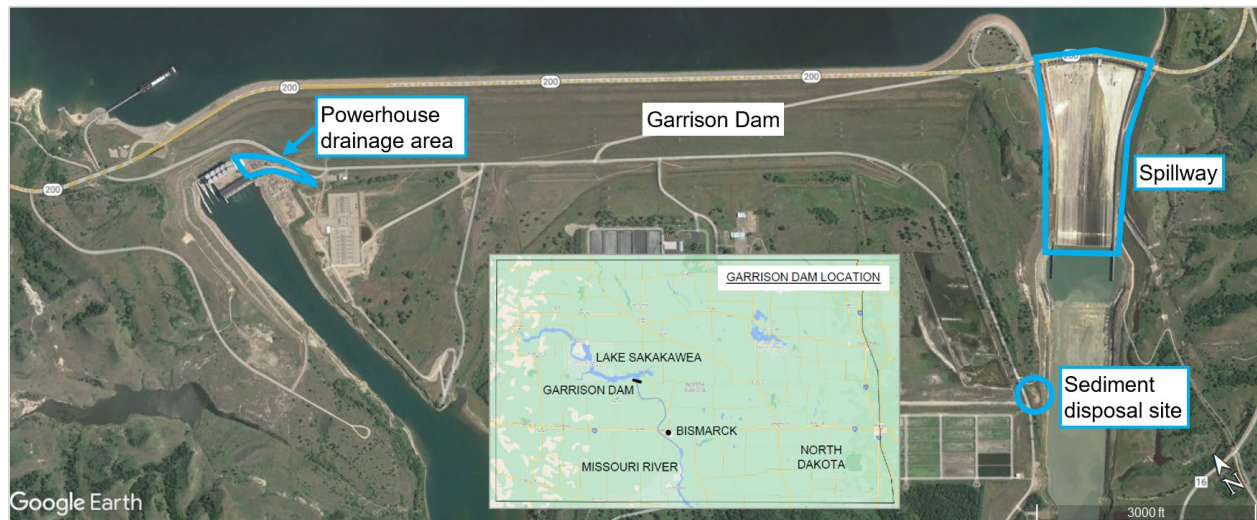


Figure 1. Garrison Dam overview map.

#### 1.1.1 Garrison Dam Background

The Garrison Dam – Lake Sakakawea Project was authorized by the Flood Control Act approved 22 December 1944, as part of the comprehensive plan for flood control and other purposes on the Missouri River Basin. The authorized purposes include flood risk management, hydroelectric power, water supply, water quality control, irrigation, recreation, navigation, and fish and wildlife. The dam was constructed primarily between 1947 and 1954. Major project components include the earth-fill dam embankment, hydroelectric generating plant, and the concrete spillway.

In 2011, the Missouri River Mainstem Reservoir System experienced record runoff and inflow into reservoirs that led to the inaugural use of the Garrison emergency spillway for flood control purposes. Issues with spillway chute underdrain system were observed after multiple manhole covers were removed from the spillway chute slab by flow during spillway operation.

Post-flood inspections, repairs, risk assessments, and studies resulted in approval to initiate design of a Garrison Dam Spillway Modification in 2023. The spillway modification design is intended to include abandonment of the existing subdrainage system and construction of a replacement subdrainage system.

## 1.2 **Drainage Systems Description**

An overview map of the spillway drainage systems is shown in Figure 2. Additional details are provided in Attachment 1.

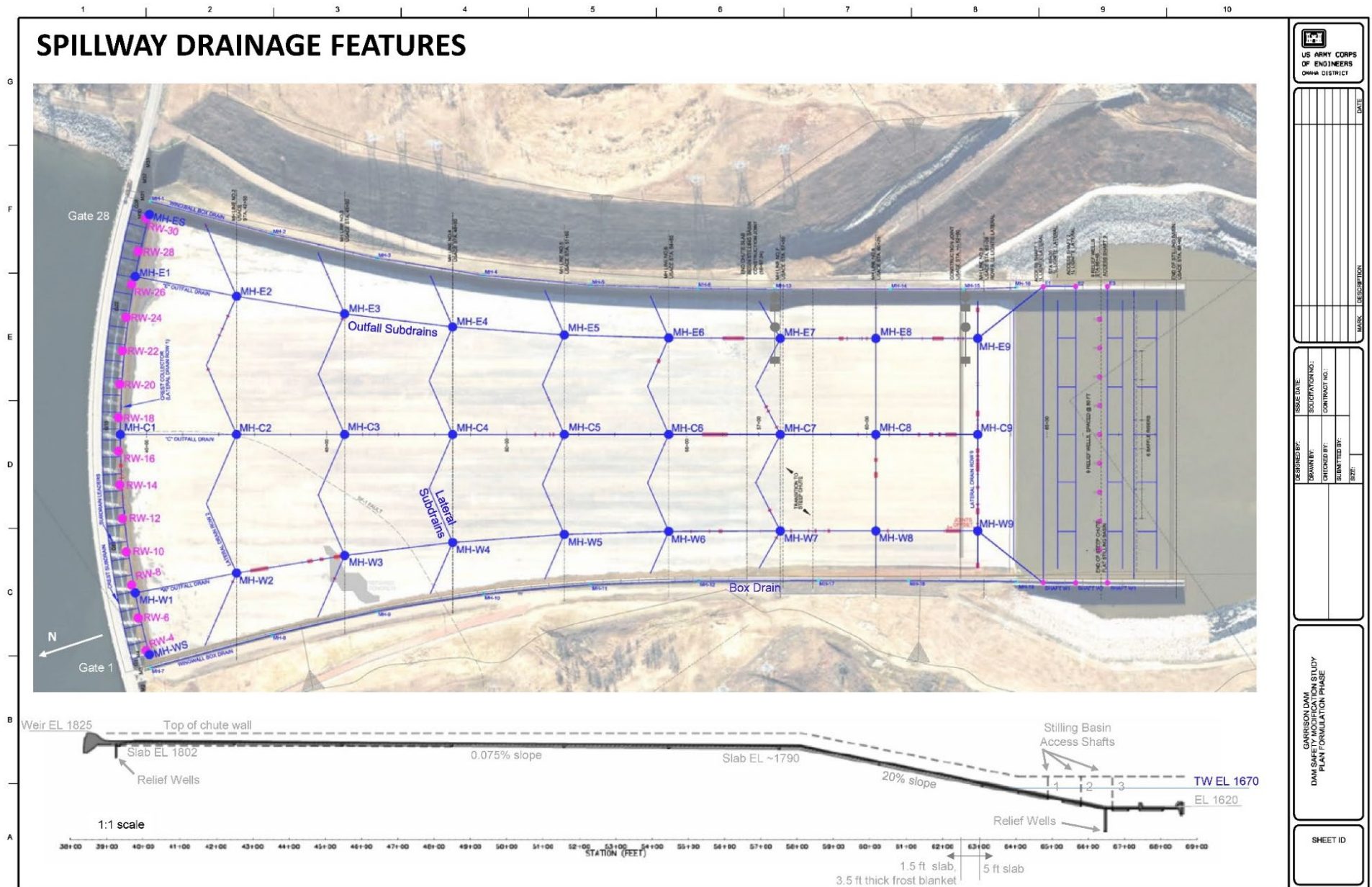


Figure 2: Subdrain and Box Drain Layout

### 1.2.1 Chute slab subdrains, general

The subdrainage pipe network is accessed from 29 manholes located within the spillway chute. The manholes are on a network of Rows 1 through 9 and outfalls “W”, “C”, and “E”. Row 1 has two additional manholes labeled MH-WS and MH-ES. The network of vitrified clay pipe segments between the manholes are listed in Table 1.

*Table 1: Chute Subdrain Lengths*

| From  | To      | Pipe Diameter (Inches) | Approx. Pipe Length (Feet) | Pipe Type          | “Row 1” |
|-------|---------|------------------------|----------------------------|--------------------|---------|
| MH-W1 | MH-WS   | 12                     | 175                        | VC Perforated      | X       |
| MH-W1 | MH-W2   | 12                     | 285                        | VC Perforated      | X       |
| MH-W2 | MH-W2E  | 12                     | 208                        | VC Perforated      |         |
| MH-W2 | MH-W2W  | 12                     | 220                        | VC Perforated      |         |
| MH-W2 | MH-W3   | 12                     | 305                        | VC Perforated/CIPP |         |
| MH-W3 | MH-W3E  | 12                     | 182.5                      | VC Perforated/CIPP |         |
| MH-W3 | MH-W3W  | 12                     | 182                        | VC Perforated      |         |
| MH-W3 | MH-W4   | 15                     | 300                        | VC Perforated      |         |
| MH-W4 | MH-W4E  | 12                     | 163                        | VC Perforated      |         |
| MH-W4 | MH-W4W  | 12                     | 151                        | VC Perforated      |         |
| MH-W4 | MH-W5   | 15                     | 310                        | VC Perforated      |         |
| MH-W5 | MH-W5E  | 12                     | 151                        | VC Perforated      |         |
| MH-W5 | MH-W5W  | 12                     | 132                        | VC Perforated      |         |
| MH-W5 | MH-W6   | 18                     | 290                        | VC Perforated      |         |
| MH-W6 | MH-W6E  | 12                     | 146                        | VC Perforated      |         |
| MH-W6 | MH-W6W  | 12                     | 132                        | VC Perforated      |         |
| MH-W6 | MH-W7   | 18                     | 302                        | VC Perforated      |         |
| MH-W7 | MH-W7E  | 12                     | 146                        | VC Perforated/CIPP |         |
| MH-W7 | MH-W7W  | 12                     | 122.5                      | VC Perforated      |         |
| MH-W7 | MH-W8   | 18                     | 275.5                      | VC Perforated/CIPP |         |
| MH-W8 | MH-W8W  | 12                     | 121.5                      | VC Perforated      |         |
| MH-W8 | MH-W9   | 18                     | 282.5                      | VC Perforated/CIPP |         |
| MH-W9 | MH-W9W  | 12                     | 121.5                      | VC Perforated      |         |
| MH-W9 | MH-W9S* | 21                     | 229                        | VC Perforated      |         |
|       |         |                        |                            |                    |         |
| MH-C1 | MH-E1   | 12                     | 440                        | VC Perforated      | X       |
| MH-C1 | MH-W1   | 12                     | 440                        | CIPP               | X       |
| MH-C1 | MH-C2   | 12                     | 320                        | VC Perforated      | X       |
| MH-C2 | MH-C2E  | 12                     | 209                        | VC Perforated      |         |
| MH-C2 | MH-C2W  | 12                     | 209                        | VC Perforated      |         |
| MH-C2 | MH-C3   | 12                     | 300                        | VC Perforated      |         |
| MH-C3 | MH-C3E  | 12                     | 182.5                      | VC Perforated      |         |
| MH-C3 | MH-C3W  | 12                     | 182.5                      | VC Perforated/CIPP |         |
| MH-C3 | MH-C4   | 15                     | 300                        | VC Perforated      |         |
| MH-C4 | MH-C4E  | 12                     | 162.5                      | VC Perforated      |         |
| MH-C4 | MH-C4W  | 12                     | 162.5                      | VC Perforated      |         |
| MH-C4 | MH-C5   | 15                     | 310                        | VC Perforated      |         |
| MH-C5 | MH-C5E  | 12                     | 151                        | VC Perforated      |         |
| MH-C5 | MH-C5W  | 12                     | 151                        | VC Perforated/CIPP |         |
| MH-C5 | MH-C6   | 18                     | 290                        | VC Perforated      |         |
| MH-C6 | MH-C6E  | 12                     | 146                        | VC Perforated      |         |
| MH-C6 | MH-C6W  | 12                     | 146                        | VC Perforated      |         |
| MH-C6 | MH-C7   | 18                     | 301                        | VC Perforated/CIPP |         |
| MH-C7 | MH-C7E  | 12                     | 146                        | VC Perforated      |         |
| MH-C7 | MH-C7W  | 12                     | 146                        | VC Perforated/CIPP |         |
| MH-C7 | MH-C8   | 18                     | 276                        | VC Perforated      |         |
| MH-C8 | MH-E8   | 12                     | 265                        | VC Perforated      |         |



| From                                     | To      | Pipe Diameter (Inches) | Approx. Pipe Length (Feet) | Pipe Type          | “Row 1” |
|------------------------------------------|---------|------------------------|----------------------------|--------------------|---------|
| MH-C8                                    | MH-W8   | 12                     | 265                        | VC Perforated      |         |
| MH-C8                                    | MH-C9   | 18                     | 283                        | VC Perforated/CIPP |         |
| MH-C9                                    | MH-E9   | 12                     | 265                        | VC Perforated      |         |
| MH-C9                                    | MH-W9   | 12                     | 265                        | VC Perforated/CIPP |         |
|                                          |         |                        |                            |                    |         |
| MH-E1                                    | MH-ES   | 12                     | 175                        | VC Perforated      | X       |
| MH-E1                                    | MH-E2   | 12                     | 285                        | VC Perforated      | X       |
| MH-E2                                    | MH-E2E  | 12                     | 220                        | VC Perforated      |         |
| MH-E2                                    | MH-E2W  | 12                     | 208                        | VC Perforated      |         |
| MH-E2                                    | MH-E3   | 12                     | 305                        | VC Perforated      |         |
| MH-E3                                    | MH-E3E  | 12                     | 182                        | VC Perforated      |         |
| MH-E3                                    | MH-E3W  | 12                     | 182.5                      | VC Perforated      |         |
| MH-E3                                    | MH-E4   | 15                     | 300                        | VC Perforated      |         |
| MH-E4                                    | MH-E4E  | 12                     | 151                        | VC Perforated      |         |
| MH-E4                                    | MH-E4W  | 12                     | 163                        | VC Perforated      |         |
| MH-E4                                    | MH-E5   | 15                     | 310                        | VC Perforated      |         |
| MH-E5                                    | MH-E5E  | 12                     | 132                        | VC Perforated      |         |
| MH-E5                                    | MH-E5W  | 12                     | 151                        | VC Perforated      |         |
| MH-E5                                    | MH-E6   | 18                     | 290                        | VC Perforated      |         |
| MH-E6                                    | MH-E6E  | 12                     | 132                        | VC Perforated      |         |
| MH-E6                                    | MH-E6W  | 12                     | 146                        | VC Perforated      |         |
| MH-E6                                    | MH-E7   | 18                     | 302                        | VC Perforated/CIPP |         |
| MH-E7                                    | MH-E7E  | 12                     | 122.5                      | VC Perforated      |         |
| MH-E7                                    | MH-E7W  | 12                     | 146                        | VC Perforated      |         |
| MH-E7                                    | MH-E8   | 18                     | 275.5                      | VC Perforated      |         |
| MH-E8                                    | MH-E8E  | 12                     | 121.5                      | VC Perforated      |         |
| MH-E8                                    | MH-E9   | 18                     | 282.5                      | VC Perforated/CIPP |         |
| MH-E9                                    | MH-E9E  | 12                     | 121.5                      | VC Perforated      |         |
| MH-E9                                    | MH-E9S* | 21                     | 229                        | VC Perforated      |         |
| <b>Total Pipe Length to be Inspected</b> |         |                        |                            | 16,180 feet        |         |
| <b>Task 1 – “Rows 2-9” Length</b>        |         |                        |                            | 14,060 feet        |         |
| <b>Task 2 - “Row 1” Length</b>           |         |                        |                            | 2,120 feet         |         |

\*Diagonal outfalls from manholes W9 and E9 cross beneath the tailwater; inspect down to the tailwater level.

### 1.2.2 Chute slab subdrains, Row 1 Subdrains

The Row 1 drainage pipes have several specific details relating to pipe inspections including:

- Leader pipes from the crest drainage and relief wells outfall into the Row 1 lateral pipe.
- The row 1 lateral pipes are anticipated to contain the greatest volumes of sediment observed in the subdrain system, with most of the sediment being drainage fill material (sand and gravel, aggregate sizes up to 3 inches). Without flushing, the sediment has previously proven to be a challenge for navigating a camera in the drains.
- As a result of the long-term movement of the spillway crest and slab, the row 1 drainage alignment is a low area behind the gates which ponds several inches of surface water. Additional coordination with the project office is required to access the five Row 1 manholes, which must be sandbagged to prevent inflow of surface water. Additionally, several of these manholes are no longer freely draining, and will need to be pumped to access the pipes.

### 1.2.3 Box Drains

The box drains are a square concrete conduit along the outside of each spillway chute wall which collect seepage through perforated vitrified clay plates. The box drain is accessible from manholes along the outside of the chute walls.

Table 2: Box Drain Segment Lengths

| From                                     | To    | Pipe Diameter (Inches) | Pipe Length (Feet) | Pipe Type          |
|------------------------------------------|-------|------------------------|--------------------|--------------------|
| <b>EAST WALL</b>                         |       |                        |                    |                    |
| MH-1                                     | MH-2  | 24                     | 340                | Concrete box drain |
| MH-2                                     | MH-3  | 24                     | 295                | Concrete box drain |
| MH-3                                     | MH-4  | 24                     | 295                | Concrete box drain |
| MH-4                                     | MH-5  | 24                     | 295                | Concrete box drain |
| MH-5                                     | MH-6  | 24                     | 295                | Concrete box drain |
| MH-6                                     | MH-13 | 24                     | 210                | Concrete box drain |
| MH-13                                    | MH-14 | 24                     | 320                | Concrete box drain |
| MH-14                                    | MH-15 | 24                     | 205                | Concrete box drain |
| MH-15                                    | MH-16 | 24                     | 145                | Concrete box drain |
| <b>WEST WALL</b>                         |       |                        |                    |                    |
| MH-7                                     | MH-8  | 24                     | 340                | Concrete box drain |
| MH-8                                     | MH-9  | 24                     | 295                | Concrete box drain |
| MH-9                                     | MH-10 | 24                     | 295                | Concrete box drain |
| MH-10                                    | MH-11 | 24                     | 295                | Concrete box drain |
| MH-11                                    | MH-12 | 24                     | 295                | Concrete box drain |
| MH-12                                    | MH-17 | 24                     | 330                | Concrete box drain |
| MH-17                                    | MH-18 | 24                     | 250                | Concrete box drain |
| MH-18                                    | MH-19 | 24                     | 300                | Concrete box drain |
| <b>Total Pipe Length to be Inspected</b> |       |                        | <b>4800 Feet</b>   |                    |

#### 1.2.4 Crest Relief Wells

The crest structure relief wells are vertical drainage features with riser pipes accessible from the chute slab surface near the spillway crest. The wells also have short horizontal outfall segments which exit into the Row 1 lateral drainage pipe. The depth of the wells varies between roughly 30 and 60 ft. The well diameter varies, beginning as a 12" formed hole in the chute slab, transitioning to a 6" cast iron pipe through the subdrainage fill, then a 6" wood casing through the foundation, with a slotted segment in a specific reach for seepage collection.

Table 3: Crest Relief Wells

| Relief Well   | Location    | Depth (feet)    |
|---------------|-------------|-----------------|
| SW 4          | Monolith 4  | ~30             |
| SW 6          | Monolith 6  | 32.9            |
| SW 8          | Monolith 8  | 34.6            |
| SW 10         | Monolith 10 | 39.5            |
| SW 12         | Monolith 12 | 41              |
| SW 14         | Monolith 14 | 43.4            |
| SW 16         | Monolith 16 | ~48             |
| SW 18         | Monolith 18 | ~35             |
| SW 20         | Monolith 20 | ~32             |
| SW 22         | Monolith 22 | ~32             |
| SW 24         | Monolith 24 | 32.3            |
| SW 26         | Monolith 26 | ~32             |
| SW 28         | Monolith 28 | 58.2            |
| SW 30         | Monolith 30 | ~55             |
| <b>Total:</b> |             | <b>550 feet</b> |

### 1.2.5 Powerplant Underseepage Drain Network

The powerplant under seepage drains provide drainage to the powerhouse area slopes and paved areas. This network includes catch basins with solid pipe segments for collecting surface runoff, and also perforated pipe segments for the purpose of collecting seepage from the dam foundation.

*Table 4: Powerplant Drain Lengths*

| From                                                   | To           | Pipe diameter<br>(inches) | Pipe length<br>(feet) | Pipe type                                                        |
|--------------------------------------------------------|--------------|---------------------------|-----------------------|------------------------------------------------------------------|
| <b>Underseepage collection drainage lines, to MH17</b> |              |                           |                       |                                                                  |
| MH11                                                   | MH12         | 12"                       | 270                   | Slotted PVC (80 ft from MH11) /<br>Perf V. C. (190 ft from MH12) |
| MH12                                                   | MH13         | 12"                       | 85                    | Perf V. C.                                                       |
| OR29R                                                  | MH29         | 8"                        | 110                   | Stainless Steel Slotted Pipe                                     |
| OR29L                                                  | MH 29        | 8"                        | 50                    | Stainless Steel Slotted Pipe                                     |
| MH29                                                   | MH13         | 12"                       | 10                    | PVC                                                              |
| OR14R                                                  | MH14         | 12"                       | 125                   | Slotted PVC                                                      |
| MH14                                                   | MH13         | 12"                       | 170                   | Slotted PVC                                                      |
| MH13                                                   | MH17         | 12"                       | 135                   | Perf V. C.                                                       |
| MH25                                                   | MH17         | 30"                       | 180                   | RCP                                                              |
| MH17                                                   | MH 18        | 36"                       | 195                   | RCP                                                              |
| <b>Underseepage collection drainage lines, to MH16</b> |              |                           |                       |                                                                  |
| MH17                                                   | MH16         | 12"                       | 150                   | Slotted PVC                                                      |
| MH15                                                   | MH16         | 12"                       | 100                   | Perf V. C.                                                       |
| MH28R                                                  | MH 27R       | 12"                       | 85                    | Solid PVC                                                        |
| MH27R                                                  | MH26R        | 12"                       | 75                    | Solid PVC                                                        |
| MH 26R                                                 | MH16         | 12"                       | 135                   | Slotted PVC                                                      |
| MH16                                                   | MH18         | 30"                       | 155                   | RCP                                                              |
| <b>Catch basin runoff drainage lines</b>               |              |                           |                       |                                                                  |
| CB10                                                   | MH25         | 12"                       | 20                    | RCP                                                              |
| CB11                                                   | MH25         | 12"                       | 75                    | RCP                                                              |
| CB21                                                   | MH25         | 12"                       | 50                    | RCP                                                              |
| CB15                                                   | MH22         | 12"                       | 30                    | RCP                                                              |
| CB17                                                   | MH22         | 12"                       | 50                    | RCP                                                              |
| CB18                                                   | MH22         | 12"                       | 50                    | RCP                                                              |
| CB19                                                   | MH22         | 12"                       | 100                   | RCP                                                              |
| MH22                                                   | MH24         | 18"                       | 160                   | RCP                                                              |
| CB13                                                   | MH23         | 12"                       | 30                    | RCP                                                              |
| CB16                                                   | MH23         | 12"                       | 30                    | RCP                                                              |
| MH23                                                   | MH24         | 12"                       | 65                    | RCP                                                              |
| CB12                                                   | MH24         | 12"                       | 30                    | RCP                                                              |
| CB14                                                   | MH24         | 12"                       | 65                    | RCP                                                              |
| CB20                                                   | MH24         | 12"                       | 110                   | RCP                                                              |
| MH24                                                   | MH25         | 24"                       | 180                   | RCP                                                              |
| CB1                                                    | MH15         | 18"                       | 245                   | RCP                                                              |
| CB2                                                    | MH17         | 12"                       | 45                    | RCP                                                              |
| CB3                                                    | MH16         | 18"                       | 20                    | RCP                                                              |
| <b>Outfall area, catch basin lines</b>                 |              |                           |                       |                                                                  |
| CB4                                                    | MH18         | 18"                       | 115                   | RCP                                                              |
| CB8                                                    | MH18         | 12"                       | 10                    | RCP                                                              |
| CB8                                                    | CB9          | 12"                       | 85                    | RCP                                                              |
| CB4                                                    | CB5          | 12"                       | 160                   | RCP                                                              |
| CB5                                                    | Upstream leg | 12"                       | 160                   | RCP                                                              |
| CB5                                                    | MH19         | 12"                       | 20                    | RCP                                                              |
| CB5                                                    | CB6          | 12"                       | 260                   | RCP                                                              |
| CB6                                                    | MH 20        | 12"                       | 10                    | RCP                                                              |
| CB6                                                    | CB7          | 12"                       | 250                   | RCP                                                              |
| CB7                                                    | MH21         | 12"                       | 35                    | RCP                                                              |
| CB7                                                    | End section  | 12"                       | 50                    | RCP                                                              |
| <b>Outfall area, main collector pipe</b>               |              |                           |                       |                                                                  |
| MH18                                                   | MH19         | 36"                       | 220                   | RCP                                                              |
| MH19                                                   | MH20         | 36"                       | 220                   | RCP                                                              |
| MH20                                                   | MH 21        | 36"                       | 265                   | RCP                                                              |
| MH 21                                                  | OUTFALL      | 36"                       | 50                    | RCP                                                              |
| <b>Total pipe length to be inspected</b>               |              |                           | <b>5,295 feet</b>     |                                                                  |

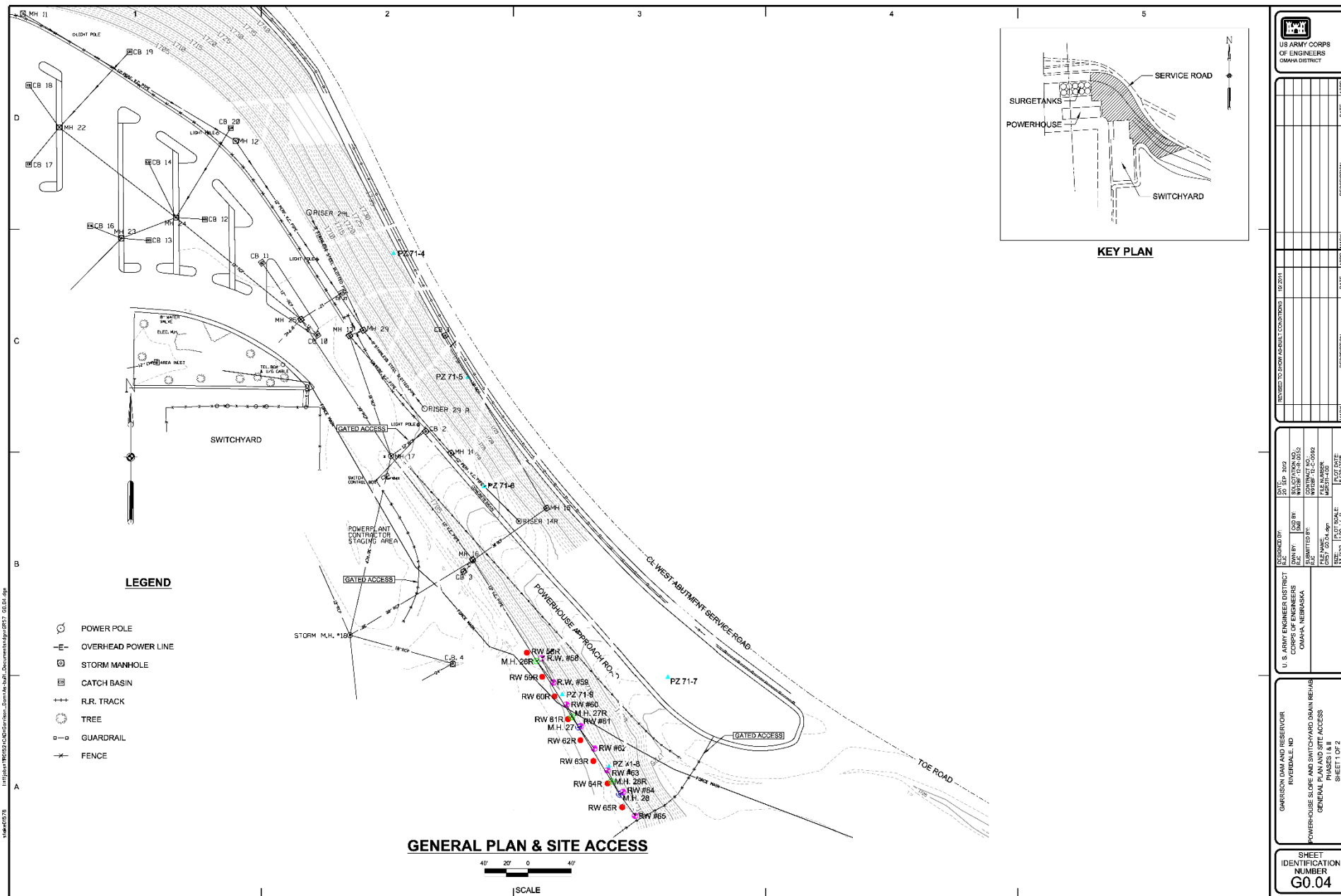


Figure 3: Powerplant Under Seepage Drain Network



### 1.3 Site Access

Spillway Access. There are gravel roads connecting to Highway 200 that run along the chute on the East and West sides of the spillway chute (Figure 4). The spillway can be accessed via the ladders on the East and West sides of the spillway near the crest. Because of restricted access the spillway is designated as a confined space area. Access requires HEC clearance and daily lock-out tag-out sign on/off. Necessary permits and safety procedures need to be followed. The spillway slab is NOT accessible without signing onto a clearance at the powerplant. A minimum of 2 weeks notice must be given to Garrison Operations in advance of work to complete clearance paperwork. Construction of an access stairway would eliminate the “limited or restricted access” resulting in the general spillway space no longer fitting the definition of confined space. HEC lock-out tag-out would still be applicable.

Manholes access. In order to enter the manholes in both the spillway and the powerplant parking lot, fall protection, ventilation, and air monitoring equipment will have to be used and provided by the Contractor. Manholes remain confined spaces regardless of the presence of an access staircase into the spillway. An activity hazard analysis (AHA) and accident prevention plan (APP), including fall protection plan and confined space plan, are required submittals for Government review prior to the execution of the work. See Sections 4.5.3 and 0 for additional information

Spillway Row 1 manholes. The row 1 manholes are in a low area which is inundated by shallow standing water. Therefore, to access the manhole it must be ring-diked by sandbags. The water from inside the ring dike can then be bailed out to allow for access to the manhole without the influence of water from the surface of the spillway.

Powerplant drains. The powerplant underseepage drainage network can be accessed via the Garrison Dam powerplant parking lot. There is gated access with intercom connection to powerplant staff. The Contractor shall provide 2 weeks notice to Garrison Operations prior to executing work near the powerplant. Coordination with Garrison Operations is required prior to work to develop a plan for powerplant personnel parking to ensure no interference with the contractor’s work.



Figure 4: Spillway site access map



Figure 5. View of spillway crest from the west abutment.



Figure 6. View facing the west spillway wall from the east.

## **2. General Contract Information**

### **2.1 Period of Performance**

The Contractor shall complete all work within 225 calendar days from issuance of the award. The Contractor shall efficiently plan, coordinate, and execute work so that materials can be acquired and all activities completed within this duration. The breakdown of dates for individual submittals and milestones is presented in Table 5 (Section 4.1). Fieldwork is not permitted within general winter conditions dates of Nov 1 to Apr 15 due to difficulty of accessing the work areas within these times.

### **2.2 Government Furnished Items and Services**

The Contractor is to provide a minimum 2 weeks notice prior to mobilizing on site to allow for these preparations. Any support required from Garrison Operations is only available between 0800 to 1500 on weekdays (Monday to Friday).

Items and support provided by the Government are to include:

- a) Water Supply. A raw (non-potable) water fire hydrant is available at the Garrison project, outside maintenance shop located at the toe of the dam. Within the spillway area, water may be taken from the stilling basin and outlet channel if approved in the APP.
- b) Sediment disposal area. Sediment collected from the drains shall be captured and properly disposed of so as not to impact Waters of the United States. The disposal site is accessible from the dam toe road as shown in Figure 1. It is located above the Ordinary High Water Mark.
- c) Prior to the work, USACE will complete a maintenance cleaning of the spillway slab surface in the Row 1 area to reduce the volume of sediment which can obscure the manholes and relief wells locations. Additionally, USACE will place markers indicating the location of the Row 1 manholes and relief wells.
- d) Confined space and Hazardous Energy Controls (HEC) briefing and daily lock-out tag-out coordination for spillway chute access. The Contractor is to report to the Garrison Powerplant Operations Control Room prior to work each day. ~~to sign on and at the end of each day to sign off (powerhouse is staffed 24 hours).~~ The Contractor shall provide their own locks for the clearance.
- e) Equipment to open specific manholes at the powerhouse site (a handheld hook).
- f) Welding. The relief well covers are currently welded shut. Assume that the welds will be removed prior to Contractor mobilization. The Government will re-weld these covers after Contractor de-mobilization. If any issues are encountered regarding welds on the relief well covers, contact the COR.

### **2.3 Contractor Furnished Items and Services**

Except for the items specifically listed in Section 2.2 above, the Contractor shall furnish all supplies, equipment, facilities and services required to complete all tasks and meet all requirements presented in the PWS, with no additional assistance from the Government. For clarity, this includes:

- a) Electrical / power. There is no electrical hook-up in the location of the camera inspections. The contractor shall supply all necessary power to the work area. If the Contractor elects to use a portable generator, KTR shall provide a spill containment plan (included in their EPP, see Section 4.5.5) to ensure no fuel enters any Government outfalls.
- b) Equipment for access, fall protection and rescue, ventilation, and air monitoring for drainage manholes access.



- c) Equipment and staffing to dewater the Row 1 manholes for access to the subdrain pipes.
- d) Crane services for lowering equipment onto the spillway slab.
- e) Typical tools for removing and re-installing manhole covers. Assume a variety of bolt sizes including 5/8". Re-fasten manhole covers with each bolt torqued to 20 ft-lbs. Replacement of the covers may require re-tapping threads.
- f) Sandbags and sand or other means for isolating the Row 1 manhole covers from surrounding surface water.

## **2.4 Contractor POC's**

The Contractor shall, at a minimum, provide contact information for the project manager, the alternate project manager, and the lead field person working at the dam. Contact information shall include office phone, cell phone, and e-mail address for each individual. In addition, the contractor shall provide a 24-hour phone number that can be utilized to contact the contractor's POCs outside of normal business hours. The Contractor will provide the updated contact lists anytime there is a change in personnel or contact information.

Project Manager. The Contractor shall assign a Project Manager (PM) to act as a primary point of contact (POC). The PM shall be responsible for the coordination of the work developed under this Contract. Upon award of this task order, the Contractor shall immediately advise the Contracting Officer and COR in writing of the individual designated as such.

## **2.5 Government POC's**

The Contractor shall be in contact with multiple Government POCs representing the Army Corps of Engineers, Omaha District. The Contracting Officer (KO) is the only POC with authority to direct the Contractor to perform services. The Government POCs including COR, Technical lead, Technical Points-of-contact, QA staff, Project Manager, and Garrison project staff for this contract will be provided after contract award.

## **2.6 PWS Modifications**

No person other than the Government Contracting Officer has the authority to make any changes to this contract action that impact cost or schedule. Authority from the Contracting Officer to the Contractor to make changes that impact cost or schedule will be in the form of an official, signed modification.

## **3. Description of Services**

The contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items services necessary to assess/inspect the drains/collection systems as defined in this Performance Work Statement except for those items specified as government furnished property and services. Remote operated camera equipment is required to document the existing condition of the systems. The contractor shall perform to the performance requirements and standards in this contract.

### **3.1 Task 1 - Flush, measure sediment, camera and report for Rows 2-9 spillway subdrain (14,060 ft)**

This work shall consist of flushing, measuring the sediment volume, camera-inspecting and reporting for all spillway slab subdrains accessed from Rows 2 through 9 manholes in the spillway chute slab. The subdrains of 12-18" I.D. perforated vitrified clay pipe extend between individual manholes 6 to 12 feet depth below the spillway slab. The subdrains are located 0 to 3 feet above the manhole floor. Additional manhole and subdrain piping details are provided in Attachment 1. For all pipe inspections and reports the contractor will use the established naming convention for each manhole and pipe segment, provided in Table 1 for the chute subdrains.

Manhole Access and Dewatering. In Rows 2 through 9, it is anticipated that the manholes are sufficiently dry such that pumping is not required to access the subdrains. However, if pumping is necessary, document pre-

existing water level and the pumping effort (rate, duration) utilized to achieve a dry condition for pipe inspections.

Manhole Observations. Prior to pumping, flushing, or other activities that would alter the existing condition of the pipe, document at each manhole a description the flow from pipes entering each manhole. For example, at MH-C5, a description may include simple qualitative flow descriptors, MH-C5 North - trickle, MH-C5 East - Dry, MH-C5 West - flowing. Current discharge rates from each subdrain is anticipated to be relatively small (less than 5 gpm). Base the flow description on these approximate flow rates: Dry (0 gpm), trickle (2-5 gpm), flow (>5 gpm). Note whether there is any change in the discharge from the pipes before/after the flushing (ie, change from “trickle” to “flow”, or “no change”.

#### Flushing and Sediment Measurement

- The Contractor’s Work Plan submittal will describe the intended cleaning equipment, flow rate, and water pressure, with care towards limitations for vitrified clay pipes as described in Section 4.4 and Table 6.
- Measuring the flushed sediment volumes is a requirement. The Contractor’s Work Plan submittal will describe the intended method to best capture and measure this sediment. Report sediment in units of volume, such as cubic feet. It shall be assumed that all pipes are approximately 20% filled with sediment and require hydro-flushing for complete and thorough inspection.
- Assume that Row 9 manholes have 1 ft depth of existing sediment (3 manholes \* 1 ft depth \* 4 ft width \* 4 ft height = 48 cubic ft). Measure and remove this sediment prior to flushing any new sediment into the manholes. Assume Rows 2 through 8 manholes have negligible levels of existing sediment. USACE may provide sacks to preserve representative samples of the flushed gravel at select manholes.
- Collect two representative sack samples of the flushed drain material relating to this Task. For Task 1, at least one of these samples is to be collected from Row 9 lateral segments.

Pipe Camera Inspection. This work shall consist of remote camera inspection of all spillway slab subdrain segments. Section 4.4 provides more detailed specifications for the camera equipment and methods.

Reporting. Reporting will include the conditions and findings from the flushing, sediment measurement, and camera inspection. Section 4.4 provides more detailed specifications for the reporting requirements.

Performance Requirements. Section 4.4 and Table 7 “Performance Requirements Summary” summarize the primary criteria by which the success of the work will be evaluated, referencing the “Inspection Requirements for Remote Inspections” stated in USACE Engineer Manual 1110-2-2902. Criteria include the quality of cleaning, care of jetting pressure in clay pipes, camera capabilities and lighting, and completeness of reporting.

### **3.2 Task 2 – Flush, measure sediment, camera and report for Row 1 spillway subdrain (2,120 ft)**

This work shall consist of flushing, measuring the sediment volume, camera-inspecting and reporting for all spillway slab subdrains accessed from Row 1 manholes in the spillway chute slab. The subdrains of 12-18” I.D. perforated vitrified clay pipe extend between individual manholes 6 to 12 feet depth below the spillway slab. The subdrains are located 0 to 3 feet above the manhole floor. Additional manhole and subdrain piping details are provided in Attachment 1. For all pipe inspections and reports the contractor will use the established naming convention for each manhole and pipe segment, provided in Table 1 for the chute subdrains.

The Row 1 drainage system is separated from the broader subdrain system in the price schedule due to the additional steps required for access into the Row 1 Manholes, the anticipated greater volume of sediment along the Row 1 pipes, and the presence of relief well and crest leader pipes into the Row 1 drains to inspect. For



purpose of this drainage inspection, Row 1 drain segments are defined as all segments attached to a Row 1 manhole and includes outfall segments MH-W1 to MH-W2, MH-C1 to MH-C2, and MH-E1 to MH-C2.

All steps and requirements for Task 1 (Subdrain Rows 2-9) also apply to Task 2 (Row 1), with additional details for Task 2 described below.

#### Row 1, Manhole Access and Dewatering.

- To prevent inflow of surface water and sediment into the manhole, place a sandbag ring or other means to isolate the manhole from the standing water. It takes approximately 12 sandbags to surround the manhole and provide some working area.
- It will be necessary for the Contractor to pump standing water within the Row 1 manholes to obtain access and camera-inspect in dry conditions. Document the depth to the water level from the top of the concrete spillway slab. Document the pumping rates and durations required to dewater the manhole, and to maintain dry conditions if continuous pumping is required. Assume there is approximately 5 ft of standing water in the manhole, and requiring 4 pumps of 2" diameter to accomplish the dewatering.
- Assume that Row 1 manholes have 1 ft depth of existing sediment (5 manholes \* 1 ft depth \* 4 ft width \* 4 ft height = 80 cubic ft). Measure and remove this sediment prior to flushing any new sediment into the manholes.

#### Flushing and Sediment Measurement

- It shall be assumed that all pipes are approximately 40% filled with sediment and require hydro-flushing for complete and thorough inspection. See PWS Attachment 1 for additional photos of the Row 1 existing conditions.

#### Row 1, Pipe Inspection

- Note the leader pipes from both the crest drainage and from relief wells. There are 14 relief well header pipes (numbered from SW-4 to SW-30, even numbers only) and 28 crest leader pipes (numbered L-1 through L-28) which outfall into Row 1 piping.
- At each junction of these connections to the Row 1 drain, the device will stop, the camera will pan to the connection, and a short video will be taken to evaluate if there is flow from these pipes into the Row 1 drain.

Performance Requirements. Section 4.4 and Table 7 "Performance Requirements Summary" summarize the primary criteria by which the success of the work will be evaluated, referencing the "Inspection Requirements for Remote Inspections" stated in USACE Engineer Manual 1110-2-2902. Criteria include the quality of cleaning, care of jetting pressure in clay pipes, camera capabilities and lighting, and completeness of reporting.

### **3.3 Task 3 – Flush, measure sediment, camera and report for spillway box drains (4,800 feet)**

This work shall consist of flushing, measuring the sediment volume, camera-inspecting and reporting for all box drains located along the outer base of the spillway chute walls. Each box drain consists of a 24" I.D. square concrete pipe and is located within individual manholes that extend 20 to 25 feet below ground surface. Each box drain discharges outside the spillway stilling basin above the tailwater level. Discharge rates from each box drain are relatively small (less than 10 gpm). For all pipe inspections and reports the contractor will use the established naming convention for each manhole and pipe segment, provided in Table 2 for the chute box drains.

Manhole Access and Dewatering. The box drain access covers are located along the outside of the chute and can be reached by vehicle. It is anticipated that the box drain manholes are sufficiently dry such that pumping is not

required to access the drains. However, if pumping is necessary, document the pre-existing water level and the pumping effort (rate, duration) utilized to achieve a dry condition for pipe inspections.

Manhole Observations. Prior to pumping, flushing, or other activities that would alter the existing condition of the pipe, document at each manhole a description the flow from the pipe segment above. Utilize the descriptors “Dry” (0 gpm), “trickle” (2-5 gpm), and “flow” (>5 gpm).

Flushing and Sediment Measurement. The Contractor’s Work Plan submittal will describe the intended cleaning equipment, flow rate, and water pressure, with care towards limitations for vitrified clay pipes as described in Section 4.4 and Table 6. While the box drain is primarily concrete pipe, there are spaced perforated vitrified clay pipe plates for collection of drainage from the chute wall backfill. The Contractor’s Work Plan submittal will describe the intended method to best capture and measure flushed sediment. Note any change in the discharge from the pipes before/after the flushing. It shall be assumed that all pipes are approximately 20% filled with sediment and require hydro-flushing for complete and thorough inspection.

Pipe Camera Inspection. This work shall consist of remote camera inspection of all box drain segments. Section 4.4 provides more detailed specifications for the camera equipment and methods. For the box drains, turn the camera to observe the condition of each vitrified clay pipe plate.

Reporting. Reporting will include the conditions and findings from the flushing, sediment measurement, and camera inspection. Section 4.4 provides more detailed specifications for the reporting requirements.

Performance Requirements. Section 4.4 and Table 7 “Performance Requirements Summary” summarize the primary criteria by which the success of the work will be evaluated, referencing the “Inspection Requirements for Remote Inspections” stated in USACE Engineer Manual 1110-2-2902. Criteria include the quality of cleaning, care of jetting pressure in clay pipes, camera capabilities and lighting, and completeness of reporting.

### **3.4 Task 4 – Sounding, camera and report for crest relief wells**

This work shall consist of depth-sounding, camera-inspecting and reporting of the 14 spillway crest relief wells. Camera inspection should consist of at minimum a video inspection, by a device suitable/specific to the inspection of wells, that documents the condition of the riser, screen, and base of each relief well. The contractor shall provide video inspection reports that document notable features along with typical conditions with annotated depths from a stated reference point.

For all pipe inspections and reports the contractor will use the established naming convention for each manhole and pipe segment, provided in Table 3 for the crest relief wells.

Sequencing. Dewatering and inspection of the relief wells is to occur after the Row 1 manholes and drain segment have separately been flushed and inspected.

Covers Access and Dewatering. To prevent inflow of surface water and sediment into the manhole, place a sandbag ring or other means to isolate the manhole from the standing water. Sound and record the depth to base of well or top of sediment. If the sounded depth indicates that the well screens have collapsed or substantially filled with sediment, notify the COR. To facilitate camera inspection in dry conditions, pump the standing water from within the relief wells with a low-capacity pump, such that there is no risk of collapsing the pipe by pumping vacuum.

Camera Inspection and Reporting. Camera inspection and documentation requirements specific to the vertical relief wells include:

- Condition of relief well screens (encrustation, deposition, damaged screen, etc.).
- Locations of seepage inflow into the relief well screens.
- Estimation of percent relief well screen is damaged, encrusted, plugged, etc..
- Condition of relief well riser pipes and joints including riser cap/cover as appropriate (structural damage, mechanical, corrosion, etc.)

- Continuous depth measurement from top of riser or manhole. The zero reference point shall be recorded at the time of the inspection and provided in the written documentation.
- A table indicating the depth to the outfall pipe invert, top of screen, bottom of screen, top of sediment/deposition (as applicable), bottom of relief well and any areas of concern/damage.

Performance Requirements. Section 4.4 and Table 7 “Performance Requirements Summary” summarize the primary criteria by which the success of the work will be evaluated, referencing the “Inspection Requirements for Remote Inspections” stated in USACE Engineer Manual 1110-2-2902. Criteria include the camera capabilities and lighting, and completeness of reporting.

### **3.5 Task 5 – Temporary Chute Access Staircase**

Construct a temporary staircase for ease of access into the chute for both the Contractor and USACE staff. The intent of the staircase is to provide improved site access relative to the existing ladder. Construction of an access stairway would eliminate the “limited or restricted access” resulting in the general spillway space no longer fitting the definition of confined space. HEC lock-out tag-out would still be applicable. Construction of the staircase can be sub-contracted. Meet all OSHA regulations and requirements of USACE EM 385-1-1, Safety and Occupational Health (SOH) Requirements.

### **3.6 Task 6 - All Other Work**

This work shall include costs associated with mobilization and de-mobilization of personnel and equipment, pre-fieldwork submittals, pre-fieldwork meetings, weekly fieldwork meetings, and fulfillment of any requirements presented in Section 2 “General Contract Information” and Section 4 “Contractor Requirements” that are not associated with any other specific tasks. Review the Submittal Register (Table 5) for a list of pre-fieldwork submittals. Table 7 “Performance Requirements Summary” summarizes the primary criteria by which the success of the work will be evaluated.

### **3.7 Task 7 (Option 1) – Flush, measure sediment, camera and report for power plant drainage (5,295 feet)**

This work shall consist of flushing, measuring the sediment volume, camera-inspecting and reporting for 5,295 linear feet of the power plant seepage drain network located at the power plant parking lot. The drainage system includes a combination of manholes and catch basins. The depth of manholes is comparable to within the spillway slab. For all pipe inspections and reports the contractor will use the established naming convention for each manhole and pipe segment, provided in Table 4 for the powerhouse drainage.

Manhole Access. See PWS Attachment 1 for a map of the powerhouse drainage network and measured flow rates. The drainage network can be roughly divided into two halves, with one half draining towards MH16 and the other half to MH17. Individual upstream drain segments may individually collect 10-20 gpm, with MH16 regularly receiving cumulative 50 gpm and MH17 at 75 gpm. Both of these manholes connect to a 36” RCP outfall system beginning at MH18. Given these inflows, access to drainage pipes may require pumping and diversion of drainage flows. Document pre-existing water level and the pumping effort (rate, duration) utilized to achieve a dry condition for pipe inspections.

Manhole Observations. Prior to pumping, flushing, or other activities that would alter the existing condition of the pipe, document at each manhole a description the flow from the pipe segment above. For the powerhouse, use the descriptors “dry” (0 gpm), “light flow” (<10 gpm), “flow” (10 gpm).

Flushing and Sediment Measurement. The Contractor’s Work Plan submittal will describe the intended cleaning equipment, flow rate, and water pressure, with care towards limitations for vitrified clay pipes as described in Section 4.4 and Table 6. The Contractor’s Work Plan submittal will describe the intended method to best capture and measure flushed sediment. Note any change in the discharge from the pipes before/after the flushing. It shall be assumed that all pipes are approximately 20% filled with sediment and require hydro-flushing for complete and thorough inspection.

Pipe Camera Inspection. This work shall consist of remote camera inspection of all powerhouse drainage segments. Section 4.4 provides more detailed specifications for the camera equipment and methods.

Reporting. Reporting will include the conditions and findings from the flushing, sediment measurement, and camera inspection. Section 4.4 provides more detailed specifications for the reporting requirements.

Performance Requirements. Section 4.4 and Table 7 “Performance Requirements Summary” summarize the primary criteria by which the success of the work will be evaluated, referencing the “Inspection Requirements for Remote Inspections” stated in USACE Engineer Manual 1110-2-2902. Criteria include the quality of cleaning, care of jetting pressure in clay pipes, camera capabilities and lighting, and completeness of reporting.

## 4. Contractor Requirements

### 4.1 Submittal Register and Schedule

Government quality assurance review and approval is required for all submittals. Unless otherwise noted, allow for 15 calendar days for Government review of submittals. Rejected submittals shall be resubmitted for approval. The approval of submittals shall not be construed as a complete check and will not relieve the Contractor of the responsibility of quality control and providing satisfactory services in accordance with this PWS.

#### 4.1.1 Submittal list and schedule

See table and sections below for more details about required submittals.

*Table 5: Submittal Register*

| Submittal Type            | Submittal Name                                                                                       | Schedule              | PWS Ref |
|---------------------------|------------------------------------------------------------------------------------------------------|-----------------------|---------|
| Pre-Field Work Submittals | <i>Included in Task 6 – All Other Work</i>                                                           |                       |         |
|                           | Pre-work meeting minutes – contract kickoff                                                          | 7 days after award    | 4.3     |
|                           | Work Plan / Quality control plan                                                                     | 14 days after award   | 4.2     |
|                           | Site safety and health officer                                                                       | 14 days after award   | 4.5.2   |
|                           | Activity Hazard Analysis (AHA)                                                                       | 14 days after award   | 4.5.3   |
|                           | Accident prevention plan (APP) including: fall protection plan, confined space plan, crane lift plan | 14 days after award   | 0       |
|                           | Environmental Protection Plan (EPP)                                                                  | 14 days after award   | 4.5.5   |
|                           | E-verify and security documentation                                                                  | 30 days after award   | 4.6.4   |
|                           | Pre-work meeting minutes – pre-mobilization                                                          | 30 days after award   | 4.3     |
|                           | Pre-work meeting minutes – on-site kickoff                                                           | 45 days after award   | 4.3     |
| Milestone                 | Fieldwork kickoff                                                                                    | 45 days after award   | -       |
| In-Progress Submittals    | <i>Included in Task 6 – All Other Work</i>                                                           |                       |         |
|                           | Weekly progress meeting minutes                                                                      | -                     | 4.3     |
| Milestone                 | All fieldwork complete                                                                               | 105 days after award* | -       |
| Final Submittals          | <i>Included in Inspection Tasks 1, 2, 3, 4, 7</i>                                                    |                       |         |
|                           | Draft camera inspection report                                                                       | 135 days after award* | 4.4     |
|                           | Final camera inspection report                                                                       | 165 days after award* | 4.4     |
| Milestone                 | All work complete                                                                                    | 195 days after award* |         |

\*Up to 30 additional days if including Task 7 (Option 1) – resulting in fieldwork complete at 135 days and contract duration at 225 days.

#### 4.1.2 Submittal form

Submittal form ENG Form 4025-R shall be used for transmitting the submittal. This form may be reproduced by the Contractor and shall be properly executed by filling out all the heading blank spaces and identifying each time



submitted. Special care should be exercised to ensure proper listing of the specification paragraph or sheet number of the contract drawings pertinent to the date submitted for each item. The Contractor shall provide copies for each submittal to the locations stated below.

#### 4.1.3 Means of approvals

Action Codes will be as indicated on the back of ENG Form 4025-R. The Contracting Officer's Representative will mark an Action Code on each submitted ENG Form 4025 prior to returning a copy to the Contractor. The approval of a submittal by the Contracting Officer shall not be construed as a complete check but will indicate only that the general method of prosecution and detailing is satisfactory. Approval will not release the Contractor from responsibility for any error that may exist, as the Contractor is responsible for satisfactory prosecution and detailing of all work.

All submittals related to performance of work must be approved before work begins. No work can begin without approved submittals. If a submittal is E coded (disapproved), the KTR must address comments and resubmit to the Government for approval prior to work. Resubmittals also are allowed 20 days for review.

#### 4.1.4 When not approved

Action Codes will be as indicated on the back of ENG Form 4025-R. The Contracting Officer's Representative will mark an Action Code on each submitted ENG Form 4025 prior to returning a copy to the Contractor. The approval of a submittal by the Contracting Officer shall not be construed as a complete check but will indicate only that the general method of prosecution and detailing is satisfactory. Approval will not release the Contractor from responsibility for any error that may exist, as the Contractor is responsible for satisfactory prosecution and detailing of all work.

#### 4.1.5 Submittal delivery contacts

The Contractor shall provide an electronic copy of ENG Form 4025-R with each submitted item to the COR and the Primary and Alternate Technical Points of Contacts listed in Section 8.0, Coordination.

#### 4.1.6 Submittal delivery means

Provide the submittals listed in the subparagraphs below to the Government for review. Provide electronic deliverables for the submittal packages that are electronically bookmarked. Transmit deliverables to the Government using DoD SAFE (<https://safe.apps.mil/>). Physical hard drives may be required for individual files larger than 8 gigabytes (GB). Electronic files shall be in a complete and useable format by the Government. Provide the original electronic file types when requested by the Government and at the conclusion of this contract. All documents shall be fully compatible with software programs identified in the narrative below without any changes required of the Government.

Provide electronic files meeting the following requirements:

- a) Meeting Minutes
  - a. Size: Letter, 8.5-inches by 11-inches.
  - b. File type: Microsoft Word.
  - c. Deliverable type: Portable Document File (PDF).
- b) Reports
  - a. Size: Letter, 8.5-inches by 11-inches.
  - b. File type: Microsoft Word.
  - c. Deliverable type: Portable Document File (PDF) and Microsoft Word.

## 4.2 **Work Plan, Quality Assurance and Control**

The Contractor shall implement a comprehensive quality assurance/quality control (QA/QC) program to ensure that all work is conducted in accordance with the PWS and associated requirements. This program shall cover all aspects of work covered under this PWS to include field activities, data collection and reduction, and reports. The

Contractor's Project Manager is responsible for the development and implementation of the QA/QC program. A QA/QC Plan will be included in the Work Plan as indicated below.

The Work Plan submittal is to include the Inspector Operator Qualifications, baseline schedule, description of cleaning and camera equipment; planned procedures for drainage access, safety, cleaning, and camera-inspection; template reporting documents, and QA/QC plan for all aspects of work. Include specifically which equipment will be lowered into the spillway chute, how sediment will be removed from the manholes and delivered out of the spillway chute to the disposal location. The presentation of the schedule is to include a Gantt chart presenting development, submission, and review dates for submittals; kickoff and duration for specific field activities, and final submittals.

#### **4.3 Pre-Work Meetings**

The Contractor shall participate in three pre-work meetings for different phases of the project following award of the requirement and prior to commencement of major work activities to coordinate with the COR and Technical Points of Contact. The Contractor is required to submit meeting minutes for each meeting.

- a) Contract kickoff meeting: Prior to commencing any activities, the Contractor shall participate in a virtual Kickoff Meeting. The COR will review contract requirements as part of the meeting. The Government will host the meeting and develop the agenda. The Contractor shall take and distribute meeting minutes for the kickoff. This meeting will cover topics including work requirements, scheduling, submittal process, and material procurement lead times. The Contract Kickoff meeting will be planned for four (4) hours.
- b) Pre-mobilization meeting: This virtual meeting shall be conducted two weeks prior to mobilization to the site. This meeting ensures all necessary on-site coordination has been completed and include review of the pre-submittals. The Pre-Mobilization meeting will be planned for two (2) hours.
- c) On-site kickoff meeting; Following mobilization and prior to the start of drilling, a formal on-site meeting will be held with Garrison Dam project staff and QA staff, the Contractor, and QC staff. The Government and Contractor shall review the QCP and plan for ongoing communications. Government and KTR will review status of all submittals to ensure they are approved before work begins during this meeting. KTR will present QCP/APP/EPP/schedule to the Government for full understanding of work. The Contractor shall document and distribute meeting minutes. The On-Site Kickoff meeting will be planned for two (2) hours.
- d) (In-Progress Submittal) KTR shall schedule a weekly meeting with the Government COR to discuss project progress once work begins. KTR to send meeting minutes to the Government after each meeting.

#### **4.4 Performance Requirements Summary**

The Contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success.

To ensure the Government receives the services for which it contracted, the COR will only pay for services received. The Performance Requirements Summary (PRS) determines whether the Contractor meets the performance standards set forth in the PWS and lists guidelines for how and when surveillance will be performed. It ensures timeliness and effectiveness and ensures that the Contractor is delivering the results specified in the PWS and resulting contract. COR contract quality assurance will be performed per the PRS surveillance schedule. No partial payments will be authorized for work performed that does not meet PRS.

Surveillance. Periodic inspections by the COR will serve as the surveillance methods. If the COR discovers deficiencies or receives complaints, the COR will alert them on to the Contractor for correction.

Unacceptable Performance. When performance is determined unacceptable, as defined in the Performance Requirements Summary, the COR will inform the Contractor that performance is unacceptable and furnish written documentation by use of the Form DD 2772, Contract Discrepancy Report. By initialing and dating the form, the Contractor is only acknowledging notification of unacceptable performance. Disputes in surveillance should be referred to the Contracting Officer (KO).

The COR will notify the KO of less than acceptable performance. If any of the services do not conform to Contract requirements, the Government may require the Contractor to perform the services again in conformity with Contract requirements, at no increase in Contract amount. When the defect(s) in services cannot be corrected by re-performance, the Government may:

- Require the Contractor to take necessary action to ensure future performance conforms to Contract requirements.
- Reduce the Contract price to reflect the reduced value of the services performed.
- If the Contractor fails to promptly re-perform the services, or to take the necessary action to ensure future performance in conformity with Contract requirements, the Government may take corrective actions to mitigate risk to the government.

#### Requirements for Remote Inspections and Documentation

Inspection Requirements for Remote Inspections stated in EM 1110-2-2902 - 6.6.2.

- 6.6.2.1 - Inspector Operator Qualifications
  - The Operator is familiar with the system and the features to be inspected ahead of time so in the field key points to check are already known. For example, the KTR should be familiar with the anticipated distance from a given MH that a Relief Well inlet will be encountered, so these features are not accidentally overlooked.
  - The individual performing the inspection is required to demonstrate their qualifications by providing training and experience records. Individuals operating the remote inspection equipment must be trained and certified by the NASSCO PACP, or an organization with equivalent standards. A minimum of one year of experience with pipe inspections using the NASSCO's PACP or an equivalent industry standard is required. It is recommended that personnel demonstrate experience with levee or dam pipes. All inspections using remote equipment are required to meet the minimum visibility and operation requirements of NASSCO's PACP or an equivalent industry standard.
- 6.6.2.2 Cleaning Requirements
  - Debris, obstructions, and sediment must be cleaned to provide an unobstructed view of the pipe's interior before video or other remote inspections are conducted. CCTV inspection is adequately accomplished through shallow depths (up to four to six inches) of clear water. If the water is turbid or too deep to permit a clear view of all pipe and joint surfaces, dewatering or diversion is required.
  - Non-aggressive cleaning following the procedures and restrictions in Table 6 removes loose debris, root intrusion, ice accumulation, or biofouling, all of which reduce a pipe's hydraulic capacity and can inhibit appurtenance operation. The magnitude of these issues depends on pipe gradient, water velocity, pipe smoothness, the nature and quantity of debris, bedload material, density of vegetation in the drainage channel, and temperature, among other factors. Remote cleaning technology is recommended over manual methods to reduce risk of operator injuries.

Table 6: VCP Cleaning from Table 7-1 EM 1110-2-2902

|                     | Cleaning Methods                                                                                                                 | Max Nozzle Orientation              | Nozzle Positioning in Pipe                                         | Max Water Pressure (psi) | Flow Rate (gpm) or nozzle movement speed (fpm)        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------|--------------------------|-------------------------------------------------------|
| Vitrified Clay Pipe | Jetting; hydromechanical, rotational, power rodders, grinding type, and bucket cleaning used when jetting is not fully effective | Any angle and distance to pipe wall | Center in pipe and no restrictions on stationary nozzle dwell time | 5000                     | 125 gpm; fpm must be determined by equipment operator |

- Additional details specific to this PWS:
  - The drain line cleaning shall remove foreign materials from the lines and restore the drain to a minimum of 90 percent of the original carrying capacity, and allowing for a thorough camera inspection. Should any conditions be encountered where this requirement is not achievable, or where damage to the pipes could result from further cleaning effort, the Contractor shall immediately contact the COR. These conditions could include major blockages that prevent cleaning, broken pipes, or severe offsets. It is recommended to provide the camera videos to the COR in these instances for resolution and communication as soon as possible and while the Contractor is still on site.
- 6.6.2.3 CCTV requirements/capabilities
  - CCTV is required to have the following capabilities or features to execute and record the inspection. Additional pipe inspections using specialized equipment may be required to document specific measurements (e.g., detected offset joints or metal loss due to corrosion).
- 6.6.2.3.1. Light Source
  - Bright, high-intensity light source that travels with the camera. Ability to control the light intensity to control glare is an important feature that can improve the quality of the video images. Lighting during the inspection must be adequate to fully illuminate, but not overly illuminate, pipe joints and individual points of interest (at a right angle to the direction of travel) for an accurate assessment. Excessive lighting or an overly-adjusted camera iris can result in a flaring of the image and exaggeration of pipe joint displacement or other pipe conditions.
- 6.6.2.3.2. Camera
  - Color, high definition (720p or better) resolution camera with remote focus, zoom, pan, and tilt capability.
- 6.6.2.3.3. Video.
  - Video image digitally recorded using a current digital multimedia video format.
- 6.6.2.3.4. Measurements.
  - Footage meter with capability to record footage reading on the video at all times. Ability to record and display distance from starting point within 0.1-foot accuracy.
- 6.6.2.3.5. Travel speed.
  - Maximum travel speed through the pipe not exceeding 25 feet per minute.
- 6.6.2.3.6. Still Images.

- Ability to take still images in .jpg or .png format of all significant defects observed during the inspection.

#### Other Requirements for Camera Equipment

- All equipment shall be rugged, constructed of non-corrosive/non-hazardous material, and be designed for the intended purpose of this type of inspection work. The camera shall be equipped with high- resolution lenses capable of a clear view (subject to water clarity) of the various well and pipe components. The camera head shall be capable of remotely rotating 360 degrees and panning from side to side 90 degrees. The camera shall also be equipped with remote controlled focus, zoom and a variable intensity light source. The camera equipment shall have the capability of providing continuous depth/length measurements. The camera equipment shall have the capability of providing continuous audio and video documentation in digital format and have the ability to capture still photos of damage or problem areas for inclusion into reports. The camera equipment shall have the capability of inspecting horizontal (and vertical, as applicable) pipe diameters and conditions specified in the site-specific requirements.

#### Inspection Documentation as stated in EM 1110-2-2902 - 6.7.

- 6.7.1. General.
  - Pipe inspections must be documented either using an inspection form during a walk-through inspection similar to the one shown in Figure 6-13 or a digital program that records the pipe condition during a remote inspection. For both walkthrough and remote inspections, PACP (or equivalent) ratings must be used to rate the defects of the pipe. Below is a list of the information that is required to be obtained as part of a pipe inspection and provided in a report for each pipe:
    - Narrative report
    - Date of inspection
    - Inspector(s) name(s)
    - Nearest dam or levee as-built station number to pipe that is being inspected
    - Coordinates of the inlet and outlet locations to six-decimal accuracy, preferred in decimal format (e.g., 38.845972 instead of 38° 50' 45.5")
    - Segment length (from inside wall of adjacent manholes to nearest 0.1 foot)
    - Description of pipe purpose
    - Type of pipe (material, diameter, shape, segment length)
    - Inspection method (walk-through or remote inspection)
    - Inside diameter of pipe
    - Defect, PACP (or equivalent) rating
    - Clock position of defect relative to the position of the hour hand of a clock
    - Distance from reference (starting) end of pipe
    - Defect measurements to establish height, width, and depth, if possible
    - Picture of each defect
  - Additional details for this PWS include:
    - Continuous video and audio documentation of the condition of wells, drain pipes or instruments
    - Still photographic (exterior & interior from video tape) documentation of inspection showing representative deterioration/damage and areas of concern in each feature inspected.
    - Delineation of the location of each identified deficiency including total length in a graphic log.
    - Condition of interior and exposed exterior of wells, drain pipes or instruments (structural damage, corrosion, deposition, etc.)
    - Delineation of areas that could not be inspected due to unanticipated field conditions.
- 6.7.2. Narrative Report.



- The CCTV and/or sonar operator/inspector must provide a brief narrative report that summarizes the following for each pipe inspected: the pipe location; start and end points; conditions of the inspection; equipment used; general condition of the pipe; and specific defects with location.
- 6.7.3. Submittals.
  - A digital report must be developed and submitted, paper inspection log or digital inspection log, narrative report (Section 6.7.2.), and inspection video. The inspection video must be configured in a current digital multimedia video format. Digital files must be configured to have the ability to use all features of the CCTV player, including fast forward capability. Submission should include all raw and native data formats. The following items must be submitted after completion of all required inspection activities:
    - Clearly labeled and organized electronic inspection videos.
    - Electronic still-capture pictures and/or sonar images of significant defects.
    - Printed and/or electronic inspection form or logs.
    - Aerial map locating deficiencies with the as-built pipe stationing or local name of culvert. The map must also show the direction of CCTV camera travel.
    - A summary table showing the stationing and results of each pipe inspection rating based on NASSCO's PACP or equivalent industry standard.

Table 7 lists the primary criteria against which the Contractor performance will be evaluated. All other requirements and standards presented in the PWS also apply towards the performance evaluation.

*Table 7: Performance Requirements Summary*

| Required Services                                                                                        | Primary Performance Standards                                                                                                                                                                                                                                                                                                                                                              | Acceptable Quality Levels                                                                                                                                                                  | Surveillance                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRS #1</b><br>Flush,<br>measure sediment,<br>camera and report<br>for Rows 2-9<br>spillway subdrain   | <ul style="list-style-type: none"> <li>Care and quality of jetting/flushing, sediment measurement, and camera inspection per Sections 3 &amp; 4.4</li> <li>Promptness of communication &amp; coordination with COR for any issues encountered.</li> <li>Completeness and schedule adherence for fieldwork and final reporting per Period of Performance and Submittal Register.</li> </ul> | <ul style="list-style-type: none"> <li>Zero deviation from standard.</li> <li>Payment will not be made for incomplete portions of work or portions where standards are not met.</li> </ul> | <ul style="list-style-type: none"> <li>Periodic fieldwork inspection by COR</li> <li>COR and USACE technical POC review of submittals</li> </ul> |
| <b>PRS #2</b><br>Flush,<br>measure sediment,<br>camera and report<br>for Row 1 subdrain                  | <ul style="list-style-type: none"> <li>Same performance standards as PRS #1</li> <li>Care of manhole lid access near spillway crest per Section 3 – prevent surface runoff into the subdrain system.</li> </ul>                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |
| <b>PRS #3</b><br>Flush,<br>measure sediment,<br>camera and report for<br>spillway box drains             | <ul style="list-style-type: none"> <li>Same performance standards as PRS #1</li> </ul>                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |
| <b>PRS #4</b><br>Sounding,<br>camera and report<br>for crest relief wells                                | <ul style="list-style-type: none"> <li>Same performance standards as PRS #1</li> <li>Care of manhole lid access near spillway crest per Section 3 – prevent surface runoff into the subdrain system.</li> </ul>                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |
| <b>PRS #5</b><br>Temporary Chute<br>Access Staircase                                                     | <ul style="list-style-type: none"> <li>Completeness of crane lift plan submittal (as part of APP) and adherence to the plan during work, meeting all OSHA regulations and EM 385-1-1 requirements</li> </ul>                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |
| <b>PRS #6</b><br>All Other Work                                                                          | <ul style="list-style-type: none"> <li>Completeness and schedule adherence for pre-submittals and weekly coordination per Submittal Register.</li> <li>Mobilization, fieldwork and de-mobilization without negative impact to the work site.</li> <li>Establishment and adherence of required safety procedures.</li> </ul>                                                                | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |
| <b>PRS #7 (Option 1)</b><br>Flush,<br>measure sediment,<br>camera and report for<br>power plant drainage | <ul style="list-style-type: none"> <li>Same performance standards as PRS #1</li> </ul>                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Same quality standards as PRS #1</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>Same surveillance / review standards as PRS #1</li> </ul>                                                 |

#### 4.5 Safety, Standards and Regulations

The Contractor shall perform to the Standards in the contract and comply with all local, State, and Federal Regulations during the services performed. The Contractor is responsible for ascertaining the extent to which these regulations affect the operations resulting from this contract. The Contractor shall comply will all applicable

regulations. In the event that one regulation is more restrictive than another, the more restrictive requirement will apply. Applicable regulations include, but are not limited to the following:

- a) Code of Federal Regulations (CFR), Title 36, Chapter III, Part 327, Rules and Regulations Governing Public Use on Corps of Engineers Water Resource Development Projects EP 1165-2-316 ([http://www.publications.usace.army.mil/Portals/76/Publications/EngineerPamphlets/EP\\_1165-2-316.pdf?ver=2013-08-22-104526-670](http://www.publications.usace.army.mil/Portals/76/Publications/EngineerPamphlets/EP_1165-2-316.pdf?ver=2013-08-22-104526-670));
- b) Engineering Manual (EM) 385-1-1 (USACE – Safety and Health Requirements Manual), dated March 2024.
- c) Occupational Safety and Health Standard (29 CFR 1910) regulations related to the Management and Control of Hazardous Wastes;
- d) RCRA - Resources Conservation and Recovery Act (Primarily 40 CFR 268);
- e) TSCA - Toxic Substances and Control Act (Primarily 40 CFR 761);
- f) DOT - Department of Transportation Regulations for the Transportation of Toxic Wastes;
- g) CERCLA-Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (SUPERFUND);
- h) CWA - Clean Water Act of 1972;
- i) Safe Drinking Water Act of 1974;
- j) All other applicable Federal, State, Country and City codes, laws, standards and regulations;

#### 4.5.1 Hours of Operation

Work hours of operation should be planned during daylight hours from 0700 to 2000 local time (Central) but may be adjusted as needed upon approval of the Government. Requests for working outside of these hours shall be made in writing to the COR. Work should be planned for daylight hours; however, light plants (if needed) may be used if approved in the APP and with approval of the COR. Any required support from Garrison Operations is only available between 0800 to 1500 on weekdays (Monday to Friday).

#### 4.5.2 Site safety and health officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO shall ensure that the requirements of 29 CFR 1926.16 are met for the project. Designate one person at the project site to function as the SSHO and not have any other duties. The SSHO or an equally qualified Alternate SSHO shall be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO shall have the required training, experience, and qualifications in accordance with EM 385-1-1. If the SSHO is off-site for a period longer than 24 hours, an equally qualified alternate SSHO shall be provided and shall fulfill the same roles and responsibilities as the primary SSHO. Contractor shall use ENG Form 6282 to designate SSHO.

#### 4.5.3 Activity hazard analysis (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity shall prepare an AHA and review it with the work crew that will perform the work. AHAs shall be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer Representative. Contractors shall use ENG Form 6206 to complete AHA.

#### 4.5.4 Accident prevention plan (APP)

The Contractor shall prepare and submit a full Site-Specific APP to the government for approval that covers plans regarding hazards for the work being performed. The Contractor shall review requirements in Chapter 2 of 2024 version of EM385-1-1 and develop this plan in accordance with section 2-7. The contractor shall use form ENG Form 6293 to develop APP and all requirements based on means/methods of performing work. The contractor shall use a qualified person to prepare the site-specific Accident Prevention Plan. The contractor shall sign and submit the

APP for approval prior to the start of work onsite. Meet all OSHA regulations and requirements of USACE EM 385-1-1, Safety and Occupational Health (SOH) Requirements.

- The KTR shall submit a fall protection plan (as part of the APP) to the Government for review in the completion of this work. This plan shall include all equipment and procedures to ensure all personnel on this project are protected against fall hazards, which includes all Operations personnel performing work in the vicinity of the contractor's tasks. It shall include a fall protection plan for general spillway access and manholes, including the powerplant area.
- The KTR shall submit a confined space plan (as part of the APP) that includes all equipment and procedures to ensure all personnel on this project are protected when entering confined spaces.
- The KTR shall submit a crane lift plan (as part of the APP) for equipment access onto the chute slab. Assume that lowering equipment over the chute walls classifies as a critical lift. Note the max wall height above the slab of approximate 30 ft, not including the safety fence height along the top of the wall. All heavy equipment and materials including cranes and outriggers shall be placed a minimum of thirty (30) feet from the face of the spillway retaining wall at all times. The Contractor shall provide a crane of adequate capacity for work in this contract and all support equipment and personnel for work in this contract. The contractor is responsible for providing all rigging, supports, cribbing material, etc. The Contractor shall submit for approval the following data for the proposed portable crane: total crane weight, wheel loads, wheel position while lifting load, outrigger position and loads at rated lifting load, location of center of rotation and lift radii for all lifts and load wings, and crane and outrigger dimensions. All calculations and sketches shall be submitted to clearly show the crane has sufficient boom length, angle, and capacity for location and lifting loads. Crane manufacturer catalogue information shall be submitted to verify information and calculations. Note locations of power lines and other constraints.

All workers on this project are required to wear appropriate PPE (outlined in the APP) that can include but isn't limited to: hardhat, safety glasses, ear protection, steel-toe boots, and when working near highways or roads, high-viz vests.

The Contractor shall follow the mandatory ENG Form 6293 (Accident Prevention Plan Worksheet) and include additional safety plans that are required for the scope of work and prepare their document for approval. The contractor shall use a qualified person to prepare the site-specific Accident Prevention Plan. For Contractor awareness, there are new requirements for Site Safety and Health oversight. The contractor shall prepare the APP in accordance with the format and requirements of EM 385-1-1 and supplemented herein. The contractor shall sign and submit the APP for approval prior to the start of work onsite.

#### 4.5.5 Environmental protection plan (EPP)

The Contractor shall prepare and submit an Environmental Protection Plan in accordance with EP 200-2-3 – Environmental Compliance Guidance and Procedures. The EPP shall outline the Contractor's plan to first avoid and then minimize negative environmental impacts to the worksite and adjacent environment. The Contractor shall provide work plan details in accordance with Federal, State, and local laws that include means to protect features plans, sediment and erosion control measures, environmental monitoring procedures, and hazardous material management plans; and all other aspects of environmental protection planned by the Contractor during this project. The Contractor shall use a qualified person approved by the COR to prepare and manage the EPP.

#### 4.5.6 Hazardous toxic and radioactive waste (HTRW)

There is no known HTRW materials at this location. If HTRW is encountered, activity shall be terminated at that location and the TPOC and COR contacted. The material will be investigated, and a modification put in place to change how the work will be further performed at that location and materials properly disposed of.

#### 4.5.7 Endangered species

*Piping plover:* The project area is located within the migration path of the Northern Great Plains Piping Plover; however, 31 years of historical surveys, along with a habitat assessment, have provided negative findings. However, the following guidelines shall be followed:

- a) If work is initiated within 0.5 miles of the project footprint prior to 15-Apr, no biological surveys to determine if Piping Plovers or suitable habitat is present are required prior to the commencement of work.
- b) If work commences after Apr-15, and through Aug15, a survey via a US Fish & Wildlife Service (FWS) permitted and trained USACE-Garrison Project Biologist is required within 48-hours of project initiation. The Contractor shall provide a 7-day advance notice of project initiation so biologists can accommodate the survey requirement in a timely manner and not cause project delays. If suitable nesting and foraging habitat is located within 0.5 miles of the project site, USACE biologists will conduct periodic surveys within a 0.5-mile radius if breaks in construction are greater than 72 hours.

If, at any point, piping plovers are observed within 0.5 miles of the project area, work shall immediately cease and the COR shall be contacted. Please contact USACE-Threatened & Endangered Species Lead, Scott Sterling at 701-301-0654, for initiating surveys, questions on regulations, or reporting a sighting and/or concern. The contractor shall work with the Government to determine the path forward. If no change in fieldwork activities is needed, the COR will coordinate with the Contractor to proceed without change. If fieldwork changes are required to accomplish the work due to the endangered species, this will be coordinated via a modification to change the requirements.

*Bald and golden eagle:* Due to the project footprint being located alongside the forested Missouri River corridor, a Bald and Golden Eagle survey will be conducted by a trained USACE-Garrison Project Biologist, Scott Sterling, if the project commences between February 1st – July 15th. If an active nest is located within 660ft of the project area, USFWS will be contacted for further guidance. All work within 660ft of the active nest shall cease until further notice. The Contractor shall provide a seven (7) day advance notice of project initiation so biologists can accommodate the survey requirement in a timely manner and not cause project delays. Please contact USACE-Threatened & Endangered Species Lead, Scott Sterling at 701-301-0654, for initiating surveys, questions on regulations, or reporting a sighting and/or concern. If endangered species are encountered during fieldwork activities, stop that portion of work and notify the Contracting Officer's Representative immediately. The contractor shall work with the Government to determine the path forward. If no change in fieldwork activities is needed, the COR will coordinate with the Contractor to proceed without change. If fieldwork changes are required to accomplish the work due to the endangered species, this will be coordinated via a modification to change the requirements.

#### 4.6 **Antiterrorism/Operation Security Requirements**

##### 4.6.1 General security requirements and guidance:

The security requirements described below apply to all contract personnel (including employees of the prime Contractor ("Contractor") and all subcontractor employees) supporting the performance requirements of this contract. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared

to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as “RAMs”), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures—this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

#### 4.6.2 Physical security and access control requirements:

All contract personnel requiring physical access to a federal installation or facility shall comply with the access control procedures of that location. Contract personnel requiring unescorted access to meet contract performance requirements on a DoD installation in the US shall be vetted by the installation/facility Provost Marshal/Directorate of Emergency Services/Security Office using the National Crime Information Center-Interstate Identification Index (commonly referred to as “NCIC-III”) and Terrorist Screening Database (commonly referred to as “TSDB”). Contract personnel shall comply with all personal identity verification requirements specified in installation/facility policies and procedures.

#### 4.6.3 Suspicious activity reporting training (e.g. iWATCH, CorpsWatch, or See Something Say Something):

All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity.

#### 4.6.4 Pre-screen candidates using E-Verify program:

Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

At least 1 person that is onsite at all times must submit fingerprints to the District Security office for a background check. Once cleared, this person can escort others. Recommend at least 2 people be cleared to provide unescorted access for project duration. Confirmation of E-verify paperwork and iWATCH / CorpsWatch training for all on-site workers is a required pre-submittal to field mobilization.

#### 4.6.5 Antiterrorism (AT) Level I training

Antiterrorism (AT) Level I training: All contract personnel requiring routine access to Army installations, facilities, and controlled access areas, or requiring network access shall complete initial and annual refresher AT Level I awareness training. Online AT Level I awareness training is available at <https://jko.jten.mil/> (website subject to change).

#### 4.6.6 Escorting in classified and/or sensitive areas

Escorting in classified and/or sensitive areas: In accordance with applicable regulations, all contract personnel who do not possess the appropriate security clearance or access privileges will be escorted in areas where they may be exposed to classified information or operations, sensitive information or activities, or restricted areas.